Lecture - 12

CO4: Compare various statistical methods of study data samples

CO5: Analyze and evaluation of different sets of data using hypothesis testing.

TEST SIGNIFICANCE OF DIFFERENCES BETWEEN TWO MEANS

we have discussed Z-test for testing the hypothesis about difference of two population means under different possibility of population variances σ_1^2 and σ_2^2. Recall from there, we have pointed out that one basic difference between Z-test and t-test is that, Z-test is used when standard deviations of both populations are known and t-test is used when standard deviations of both populations are unknown. But in practice standard deviations of both populations are not known, so in real life problems t-test is more suitable compared to Z-test.

Assumptions

This test works under following assumptions:

(i) The characteristic under study follows normal distribution in both the populations. In other words, both populations from which random samples are drawn should be normal with respect to the characteristic of interest.

(ii) Samples and their observations both are independent to each other.

(iii) Population variances σ_1^2 and σ_2^2 are both unknown but equal.

(iv) For describing this test, let there be two normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ under study. And we have to draw two independent random samples, say, $X_1, X_2, X_3, \dots, X_{n_1}$ *and* $Y_1, Y_2, Y_3, \dots, Y_{n_2}$ of sizes n_1 and n_2 from these normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively. Let $\bar{X}$ and $\bar{Y}$ be the means of first and second sample respectively. Further, suppose the variances of both the populations are unknown but are equal, i.e., $\sigma_1^2 = \sigma_2^2 = \sigma^2$ (say). In this case, σ^2 is estimated by value of pooled sample variance S_p^2 where,

$$S_P^2 = \frac{1}{n_1 - n_2 - 2}[(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2]$$

And

$$S_1^2 = \frac{1}{n_1 - 1}\sum_{i=1}^{n_1}(X_i - \bar{X})^2, \quad S_2^2 = \frac{1}{n_2 - 1}\sum_{i=1}^{n_2}(Y_i - \bar{Y})^2,$$

This can also be written as

$$S_P^2 = \frac{1}{n_1 - n_2 - 2}\left[\sum_{i=1}^{n_1}(X_i - \bar{X})^2 + \sum_{i=1}^{n_2}(Y_i - \bar{Y})^2\right]$$

For computational simplicity, use the following formulae for $\bar{X}, \bar{Y} and\ S_P^2$:

$$\bar{X} = a + \frac{1}{n_1}\sum d_1, \bar{Y} = b + \frac{1}{n_2}\sum d_2 \ and$$

$$S_P^2 = \frac{1}{n_1 + n_2 - 2}\left[\left\{\sum d_1^2 - \frac{(\sum d_1)^2}{n_1}\right\} + \left\{\sum d_2^2 - \frac{(\sum d_2)^2}{n_2}\right\}\right]$$

Where, $d_1 = (X - a)\ and\ d_2 = (Y - b)$, ‘a’ and ‘b’ are the assumed arbitrary values.

Now, follow the same procedure as we have discussed, that is, first of all we have to setup null and alternative hypotheses. Here, we want to test the hypothesis about the difference of two population means so we can take the null hypothesis as

$$H_0: \mu_1 = \mu_2\ \ [no\ difference\ in\ means]$$

[Here $\theta_1 = \mu_1\ and\ \theta_2 = \mu_2$ if we compare it with general procedure]

Or

$$H_0: \mu_1 - \mu_2 = 0\ \ [difference\ in\ two\ means\ is\ 0]$$

and the alternative hypothesis as

$H_1: \mu_1 \neq \mu_2$ [for two tailed test]

$$\left.\begin{array}{l} H_0: \mu_1 \leq \mu_2\ and\ H_1: \mu_1 > \mu_2 \\ H_0: \mu_1 \geq \mu_2\ and\ H_1: \mu_1 < \mu_2 \end{array}\right\}\ [for\ one-tailed\ test]$$

For testing the null hypothesis, the test statistic t is given by

$t = \frac{\bar{X} - \bar{Y}}{S_P\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t\_(n_1 + n_2 - 2)$ under H_0

After substituting values of $\bar{X}, \bar{Y}, S_P, n_1$ *and* n_2 we get calculated value of test statistic t. Then we look for critical (or cut-off or tabulated) value(s) of test statistic t from the t-table. On comparing calculated value and critical value(s), we take the decision about the null hypothesis as discussed.

Let us do some examples to become more user friendly with the test explained above.

Example 1: In a random sample of 10 pigs fed by diet A, the gain in weights (in pounds) in a certain period were

12, 8, 14, 16, 13, 12, 8, 14, 10, 9

In another random sample of 10 pigs fed by diet B, the gain in weights (in pounds) in the same period were

14, 13, 12, 15, 16, 14, 18, 17, 21, 15

Assuming that gain in the weights due to both foods follow normal distributions with equal variances, test whether diets A and B differ significantly regarding their effect on increase in weight at 5% level of significance.

Solution: Here, we can test that diets A and B differ significantly regarding their effect on increase in weight of pigs. If μ_1 and μ_2 denote the average gain in weights due to diet A and diet B respectively then our claim is $\mu_1 \neq \mu_2$ and its complement is $\mu_1 = \mu_2$. Since complement contains the equality sign so we can take the complement as the null hypothesis and the claim as the alternative hypothesis. Thus,

$$H_0: \mu_1 = \mu_2 \; and \; H_1: \mu_1 \neq \mu_2$$

Since the alternative hypothesis is two-tailed so the test is two-tailed test.

Since it is given that the increase in the weight due to both foods follow normal distributions and population variances are equal and unknown. And other assumptions of t-test for testing a hypothesis about difference of two population means also meet. So we can go for this test.

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X}-\bar{Y}}{S_P\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \sim t\_(n_1 + n_2 - 2) \quad \text{under } H_0$$

Diet A			Diet B		
X	d_1=(X-a) a=12	d_1^2	Y	d_2=(Y-b) b=16	d_2^2
12	0	0	14	-2	4
8	-4	16	13	-3	9
14	2	4	12	-4	16
16	4	16	15	-1	1
13	1	1	16	0	0
12	0	0	14	-2	4
8	-4	16	18	2	4
14	2	4	17	1	1
10	-2	4	21	5	25
9	-3	9	5	-1	1
Total	-4	70		-5	65

Here, a =12, b = 16 are assumed values.

From above calculations, we have

$$\bar{X} = a + \frac{1}{n_1}\sum d_1 = 12 + \frac{-4}{10} = 11.6$$

$$\bar{Y} = b + \frac{1}{n_2}\sum d_2 = 16 + \frac{-5}{10} = 15.5$$

$$S_P^2 = \frac{1}{n_1 + n_2 - 2}\left[\left\{\sum d_1^2 - \frac{(\sum d_1)^2}{n_1}\right\} + \left\{\sum d_2^2 - \frac{(\sum d_2)^2}{n_2}\right\}\right]$$

$$S_P^2 = \frac{1}{10 + 10 - 2}\left[\left\{70 - \frac{(-4)^2}{10}\right\} + \left\{65 - \frac{(-5)^2}{10}\right\}\right]$$

$$= \frac{1}{18}(68.4 + 62.5) = 7.27$$

$$S_P = \sqrt{7.27} = 2.70$$

For testing the null hypothesis, the test statistic t is given by

$$t = \frac{\bar{X} - \bar{Y}}{S_P\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{11.6 - 15.5}{2.70\sqrt{\frac{1}{10} + \frac{1}{10}}} = -\frac{3.90}{2.70 * 0.45} = -\frac{3.90}{1.215} = -3.21$$

The critical values of test statistic t for two-tailed test corresponding $(n_1 + n_2 - 2) = 18$ df at 5% level of significance are

$$\pm t_{(n_1+n_2-2),\alpha/2} = \pm t_{(18),0.025} = \pm 2.101$$

Since calculated value of test statistic t (= −3.21) is less than critical values (± 2.101) that means calculated value of test statistic t lies in rejection region, so we reject the null hypothesis and support the alternative hypothesis i.e. support our claim at 5% level of significance.

Thus, we conclude that samples do not provide us sufficient evidence against the claim so diets A and B differ significantly in terms of gain in weights of pigs.

Now, you can try the following exercises.

- Two different types of drugs A and B were tried on some patients for increasing their weights. Six persons were given drug A and other 7 persons were given drug B. The gain in weights (in ponds) is given below:

Drug A	5	8	7	10	9	6	-
Drug B	9	10	15	12	14	8	12

Assuming that increment in the weights due to both drugs follows normal distributions with equal variances, do the both drugs differ significantly with regard to their mean weights increment at 5% level of significance?

- To test the effect of fertilizer on wheat production, 26 plots of land with equal areas were chosen. Half of these plots were treated with fertilizer and the other half were untreated. Other conditions were the same. The mean yield of wheat on the untreated plots was 4.6 quintals with a standard deviation of 0.5 quintals, while the mean yield of the treated plots was 5.0

quintals with standard deviations of 0.3 quintals. Assuming that yields of wheat with and without fertilizer follow normal distributions with equal variances, can we conclude that there is significant improvement in wheat production due to effect of fertilizer at 1% level of significance?

- The means of two random samples of sizes 10 and 8 drawn from two normal populations are 210.40 and 208.92 respectively. The sum of squares of the deviations from their means is 26.94 and 24.50 respectively. Assuming that the populations are normal with equal variances, can samples be considered to have been drawn from normal populations having equal mean.